Question

There are six chemical equations as follows:

1. Industrially, sodium borohydride is prepared by the reaction of metallic sodium, hydrogen gas, silicon dioxide, and anhydrous borax under high temperature and pressure.

2. In analytical chemistry, the content of diallyl trisulfide in garlic is analyzed using the potassium permanganate method, with sulfuric acid providing the acidic environment.

3. Methyl-substituted isocyanic acid reacts with sodium hydroxide, and this reaction can be referenced in organic reaction mechanisms.

4. Industrially, $K_2[Fe(CN)_5(NO)] \cdot 2H_2O$ is prepared by the reaction of potassium ferrocyanide and nitric acid.

5. When iodine is added to concentrated aqueous ammonia, a black precipitate is obtained, in which the mass fraction of iodine is 92.47%.

6. Lithium aluminum hydride reacts with copper(I) iodide to form a new covalent hydride, and this reaction does not involve a redox process.

Regarding the above six equations, when the coefficients are balanced and simplified to the smallest whole number ratio, the following statements are made:

1. The sum of the product coefficients in Equation 1 is even.

2. The sum of the reactant coefficients in Equation 2 is 120.

3. Equation 3 has only two products, and the sum of their coefficients is 2.

4. One of the products in Equation 4 is explosive, and the sum of the product coefficients is 4.

5. The sum of the reactant coefficients in Equation 5 is even.

6. The product coefficients in Equation 6 include a perfect square number other than 1.

Which of the following options includes **all** the incorrect statements?

A. 1,2

B. 3,4

C. 1,4

D. 1,4,6

E. 2,4,5

F. 2,3,6

G. None of the above options are correct

Answer

Correct Answer: C

Detailed Explanation

Equation 1: Anhydrous borax is $Na_2B_4O_7$. Neither boron nor silicon atoms exhibit strong oxidizing properties, whereas elemental sodium is highly reducing. During the formation of sodium borohydride, four hydride ions are generated, suggesting that hydrogen gas is reduced by elemental sodium to hydride ions, thereby balancing the equation. At high temperatures, silicon dioxide transforms into silicate.

$$Na_2B_4O_7 + 16Na + 7SiO_2 + 8H_2 = 4NaBH_4 + 7Na_2SiO_3$$

The sum of the product coefficients is 11, an odd number, so Statement 1 is incorrect.

CHECKPOINT 0.5 PTS

Hydrogen gas is reduced by elemental sodium to hydride ions

CHECKPOINT 0.5 PTS

At high temperatures, silicon dioxide transforms into silicate

CHECKPOINT 1 PTS

Equation 1 $Na_2B_4O_7 + 16Na + 7SiO_2 + 8H_2 = 4NaBH_4 + 7Na_2SiO_3$

CHECKPOINT 0.5 PTS

The sum of the product coefficients is 11, an odd number, so Statement 1 is incorrect.

Equation 2: Under acidic conditions, potassium permanganate oxidizes to form divalent manganese sulfate, while diallyl trisulfide, as an organic compound, is completely oxidized to produce carbon dioxide and sulfate ions. This allows the equation to be balanced:

$$5(CH_2CHCH_2)_2S_3 + 52KMnO_4 + 63H_2SO_4 = 26K_2SO_4 + 52MnSO_4 + 30CO_2 + 88H_2O$$

The sum of the reactant coefficients is 120, so Statement 2 is correct.

CHECKPOINT 0.5 PTS

Under acidic conditions, potassium permanganate oxidizes to form divalent manganese sulfate

CHECKPOINT 0.5 PTS

Diallyl trisulfide is completely oxidized to produce carbon dioxide and sulfate ions

CHECKPOINT 1 PTS

$$5(CH_2CHCH_2)_2S_3 + 52KMnO_4 + 63H_2SO_4 = 26K_2SO_4 + 52MnSO_4 + 30CO_2 + 88H_2O$$

CHECKPOINT 0.5 PTS

The sum of the reactant coefficients is 120, so Statement 2 is correct.

Equation 3: Considering the organic reaction mechanism, hydroxide ions act as strong nucleophiles, and the carbon atom in the isocyanate group is electrophilic, leading to nucleophilic attack and the formation of an amide-like intermediate. Under alkaline conditions, the amide-like intermediate undergoes hydrolysis, producing carbon dioxide and a primary amine, with carbon dioxide further converting to carbonate in the alkaline environment. Thus, the equation is:

$$CH_3NCO + 2NaOH = CH_3NH_2 + Na_2CO_3$$

There are only two products, and the sum of their coefficients is 2, so Statement 3 is correct.

CHECKPOINT 1 PTS

Hydroxide ions nucleophilically attack the carbon atom in the isocyanate group to form an amide-like intermediate

CHECKPOINT 1 PTS

$CH_3NCO + 2NaOH = CH_3NH_2 + Na_2CO_3$

Equation 4: Since the product contains only five cyanide groups, if the NO in the complex is assumed to be neutral, the iron would be oxidized to the +3 oxidation state. Nitric acid is a strong oxidizer, suggesting that the -3 nitrogen in the cyanide group is oxidized to +2 in NO, while the nitrate ion itself is reduced to -3 as ammonium. The +2 carbon is also oxidized to +4, appearing as carbon dioxide. Electron calculations show that one nitrate ion can oxidize exactly one molecule of $K_4Fe(CN)_6$. Thus, the balanced equation is:

$$K_4Fe(CN)_6 + 4HNO_3 = K_2[Fe(CN)_5(NO)] \cdot 2H_2O + CO_2 + NH_4NO_3 + 2KNO_3$$

CHECKPOINT 1 PTS

One nitrate ion can oxidize exactly one molecule of $K_4Fe(CN)_6$

CHECKPOINT 1 PTS

$K_4Fe(CN)_6 + 4HNO_3 = K_2[Fe(CN)_5(NO)] \cdot 2H_2O + CO_2 + NH_4NO_3 + 2KNO_3$

Ammonium nitrate in the products is explosive, and the sum of the product coefficients is 5, so Statement 4 is incorrect.

CHECKPOINT 0.5 PTS

Ammonium nitrate in the products is explosive, and the sum of the product coefficients is 5, so Statement 4 is incorrect.

Equation 5: The mass fraction of iodine in the precipitate is 92.47%. Testing possible numbers of iodine atoms, when the count is 3, the remaining molecular weight is 31, corresponding to two nitrogen atoms and three hydrogen atoms. Thus, the precipitate's chemical formula is $NI_3 \cdot NH_3$. Clearly, iodine undergoes disproportionation, forming ammonium iodide. The equation is:

CHECKPOINT 1 PTS

The precipitate's chemical formula is $NI_3 \cdot NH_3$. Iodine undergoes disproportionation

$$5NH_3 + 3I_2 = NI_3 \cdot NH_3 + 3NH_4I$$

The sum of the reactant coefficients is 8, an even number, so Statement 5 is correct.

CHECKPOINT 1 PTS

$$5NH_3 + 3I_2 = NI_3 \cdot NH_3 + 3NH_4I$$

CHECKPOINT 0.5 PTS

The sum of the reactant coefficients is 8, an even number, so Statement 5 is correct.

Equation 6: Since no redox occurs, the newly formed covalent hydride is determined to be copper hydride, leading to the equation:

CHECKPOINT 0.5 PTS

The newly formed covalent hydride is copper hydride

$$LiAlH_4 + 4CuI = 4CuH + LiI + AlI_3$$

CHECKPOINT 1 PTS

$$LiAlH_4 + 4CuI = 4CuH + LiI + AlI_3$$

The product coefficients include 4, a perfect square, so Statement 6 is correct.

CHECKPOINT 0.5 PTS

The product coefficients include 4, a perfect square, so Statement 6 is correct.

In conclusion, statement 1 and 4 are incorrect, so option C is selected.